

Evaluation of physicochemical properties, release kinetics, and in vitro/in vivo wound healing activity of the electrospun nanofibres loaded with the natural antioxidant oil from *Pistacia atlantica*

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ABSTRACT

Herbal medicine is an extraordinary source of bioactive compounds, and developing them can aid in the wound healing process. This study focuses on the optimal production of nanofibres containing an effective herbal extract. The nanofibres were produced by electrospinning a polymer solution containing polyvinyl alcohol (PVA) and sodium alginate (ALG) with *Pistacia atlantica* oil (PAO) (0, 1.5%, and 2.5% (w/v)). The PAO is rich in different biological properties including antioxidant, anti-inflammatory and antibacterial activities. The addition of 1.5% (w/v) PAO to PVA/ALG solutions increased surface tension and conductivity by 3.5% and 3.86%, respectively. Besides, the scanning electron microscopy (SEM) images revealed that the diameter of the nanofibres increased from $191 \text{ nm} \pm 15 \text{ nm}$ – $259 \text{ nm} \pm 10 \text{ nm}$. According to differential scanning calorimetry (DSC) results, the nanofibres loaded with 1.5% (w/v) PAO exhibited higher thermal stability compared to PVA/ALG nanofibres without PAO. Based on the in vitro drug release experiments, the PAO release from nanofibres followed the Korsmeyer-Peppas model and the pseudo-Fickian mechanism. Cell culture and cytotoxicity assay results illustrated that the PAO-loaded nanofibres had non-cytotoxic effects against L-929 cells with suitable fibroblast cell proliferation and cell viability. Animal test results showed that the PAO-loaded nanofibres significantly enhanced wound closure by 92.07% compared to PAO-unloaded nanofibres ($p < 0.02$). The wound healing ability of PAO was also confirmed by the histopathological findings, with PAO increasing re-epithelization and successfully attenuating tissue inflammation. Overall, this study introduces a new type of wound dressing containing PAO, a wound healing stimulant with excellent properties such as uniform structure, antibacterial activity, and breathability. Consequently, the results suggest that the investigated nanofibres may merit being considered as an ideal wound dressing for future clinical investigations into wound therapy and regeneration.

1. Introduction

Generally, the skin may be exposed to numerous tissue injuries and wounds in daily life. The methods of treatment and wound healing depend on the type of wound [1]. Typically, wounds can be classified using various methods such as healing time, the extent of tissue loss, and affected layers [2]. The healing time of wounds varies greatly [3]; some of them are normal heal which are called acute wounds [4]. In contrast, a chronic wound may happen when the healing process does not progress normally [5]. The management of chronic wounds is a global health problem, hence 6.5 million patients in the United States suffer from chronic skin injuries [6], such as venous congestion [7], pressure sores

[8], diabetic ulcers and iatrogenic wounds [9,10]. It is estimated that 463 million people worldwide were diabetic in 2019 [11]. About a quarter of these patients have foot ulcers on the extremities [12]. Although skin recovery after injury is a natural phenomenon, the speed of healing and prevention of infection are key clinical challenges [13]. Even simple injuries require proper care and attention. The natural process of wound healing can lead to dysfunction and then failure of the healing process [14]. Incomplete wound healing causes considerable stress and discomfort to patients [15]. Using wound dressing is a successful treatment method to enhance the healing process [16,17]. Nowadays, wound dressings are categorized in various aspects based on their nature of action [18,19]. Traditional or passive dressings such as

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gauze and tulle dressings are inexpensive, have high absorbent capacity, and easily isolate the wound from contamination [20,21]. However, they adhere to the wound bed and do not improve hemostasis [22,23]. Then the wound interface dries out because moisture levels are not controlled. The disadvantages are a pain when changing the dressing and delay in the healing process [11,24].

Another type of wound dressing is interactive products [25]. Interactive wound dressings include foams [26], hydrocolloids [27] and polymeric films [28], which are usually transparent, water vapor and oxygen permeable, bacteria barrier from the external environment, and flexible to use [29,30]. It is well known that these dressings are proposed for low-exuding wounds. The last types of wound dressings are advanced therapeutic [31] or bioactive dressings [32], such as drug delivery systems [33] and biological dressings [34]. Bioactive wound dressing can contain a variety of therapeutic agents such as anti-inflammatories, antimicrobials [35], vitamins, growth factors, herbal extracts and even cells [36]. Various natural herbal extracts, such as Aloe Vera [37], *Garcinia cowa* [38], *Lawsonia inermis* (henna) [39], *Tecomella undulate* [40], *Bidens pilosa* [41], *Centella Asiatica* [42], *Azadirachta indica* [43], *Chrysin* [44], *Curcumin* [45], *Myristica andamanica* [46], *Grape seed* [47], *gum tragacanth* [48], *Fenugreek* [49], and *Pistacia atlantica* [50] can accelerate the healing process [51]. *Pistacia atlantica* consists of different parts including leaves, fruit and resin extract [52]. All parts of the plant are useful and effective pharmacologically as antidiabetic [53], anti-tumor [54], antipyretic [55], laxative [56], antiviral [57], antinociceptive [58], softeners, and wound healing agents [59]. *Pistacia atlantica* has been traditionally used to treat gastrointestinal disorders. The extracted oil from *Pistacia atlantica* has been shown to significantly reduce the intensity of gastric mucosal damage [60]. The significant wound healing properties of this plant include antioxidant, antibacterial, anti-inflammatory and antimicrobial activity [61]. The antibacterial and antifungal activities of the extracted oil from *Pistacia atlantica* can be attributed to several compounds including α -pinene, linalool, verbenone, flavones, carvacrol, and terpenoids such as mono, di-, and triterpenoids [62].

Advanced therapeutic dressings could incorporate a variety of therapeutic agents [63] and natural herbal extracts [64] into a wide range of different wound dressings constructions, including sponges, hydrogels, micro/nanofibres, films, etc. [31]. Among these, electrospun nanofibres have small pore size, high porosity, low-cost process, high flexibility, and high surface area-to-volume ratios [65] that provide efficient loading and delivery of therapeutic and bioactive molecules for wound healing and skin substitutes [66]. Among various methods, electrospinning is a powerful process for the fabrication of fibres whose diameters can be adjusted from several micrometers to tens of nanometers [67]. This method is simple, cost-effective, scalable, and useful. Due to the nanostructured architecture produced by this method, the mat may display improved biological, physical, and chemical features compared to the other methods [68]. Electrospun nanofibres can be fabricated from a variety of natural polymers [51] (e.g. chitosan, silk, fibronectin, sodium alginate, collagen, gelatin, ethyl cellulose), synthetic polymers (e.g. poly(L-lactic acid), tyrosine-derived polycarbonates, polycaprolactone, poly(ethylene oxide), polyurethane, poly(vinyl chloride), poly(vinyl pyrrolidone), polyvinyl alcohol) or combinations of these polymers [69,70]. In recent years, the use of natural polymers or biopolymers to fabricate wound dressings has exhibited attractive and diverse advantages [71], such as hydrophilicity, biodegradability, bioresorbability, non-irritation, non-toxicity, mechanical properties, and drug loading and release capabilities [35,72,73]. Among them, sodium alginate is a non-toxic polysaccharide extracted from various types of brown seaweed [74]. Sodium alginate-based wound dressings provide excellent properties in a reasonably moist environment and can absorb excess wound exudate without adhering to wounds [66,75]. However, the sodium alginate polymer solution is not easily electrospun due to the lack of chain entanglement, high electrical conductivity and high surface tension [76]. To overcome this limitation, sodium alginate is

blended with synthetic polymers such as polyvinyl alcohol (PVA), which enhances the spinnability of the natural polymer [77].

In the present study, aqueous solutions of PVA, *Pistacia atlantica* oil (PAO), and sodium alginate (ALG) were blended, and nanofibres were fabricated using the electrospinning technique for wound healing purposes. Here, the polymer solutions were characterized by conductometry, tensiometry, and rheometry experiments. The effects of PAO in the polymer solutions on the morphology, surface roughness, and thermal properties of the fabricated nanofibres were investigated using scanning electron microscopy (SEM), atomic force microscopy (AFM), and differential scanning calorimetry (DSC), respectively. The cytotoxicity of the PAO-loaded nanofibres was also evaluated by 3-(4,5-dimethyl-2-thiazolyl)-2,5-diphenyl-2H-tetrazolium bromide (MTT) viability assay. Finally, the effect of the PAO-loaded nanofibres on skin wound healing in male albino mice was assessed in vivo.

2. Materials and methods

2.1. Materials

Polyvinyl alcohol (PVA, MW of 72,000 Da), sodium alginate (SA, medium viscosity 3500 cP), Triton X-100, fetal bovine serum (FBS), penicillin, streptomycin, L-glutamine, dimethyl sulfoxide (DMSO), Dulbecco's Modified Eagle's Medium (DMEM), and 3-(4, 5-dimethylthiazol-2-yl)-2, 5-diphenyltetrazolium bromide (MTT) were obtained from Sigma-Aldrich (USA) Company. Sodium chloride, sodium hydroxide, monopotassium phosphate, potassium chloride, and disodium phosphate were also purchased from Merck Chemical Company (Germany) to make the phosphate buffer solution (PBS, pH = 7.4) [78]. Moreover, PAO was extracted from the leaves and fruits of the *Pistacia atlantica* as described in our previous work [79]. The mouse fibroblast cell line (L-929) was obtained from the Pasteur Institute of Iran to check the biocompatibility and cell adhesion of the produced nanofibres.

2.2. Preparation and characterization of working fluids

PVA/ALG mixture was prepared by dissolving PVA (10% w/v) and ALG (2% w/v) separately in deionized water at 80 °C and room temperature for 3 h, respectively. Afterwards, an 80/20 v/v mixture of the PVA and ALG polymer solution was stirred for 20 h. Then, 1% (v/v) Triton X-100 was added to this solution as a surfactant. Finally, PAO extract was added to the PVA/ALG/Triton solution to obtain solutions containing 1.5% (w/v) and 2.5% (w/v) PAO extract. The electrical conductivity of ALG solutions was determined using a CDM120 conductivity meter (MeterLab). The surface tension of solutions was measured with the LAUDA/TD1C tensiometer. All measurements were performed in triplicate at room temperature.

2.3. Electrospinning and characterization of nanofibres

The PVA/ALG-PAO nanofibres were produced by electrospinning. For the electrospinning process, the polymer solution was placed into a 5 mL plastic syringe with a 19G stainless steel needle at a constant feed rate of 0.3 mL/h. A high voltage of 18–20 kV was applied between the needle and the grounded collector with a tip-to-collector distance of 10 cm. Ambient conditions such as temperature and relative humidity were set at 40 %–45% and 25 °C–30 °C, respectively, to achieve homogeneous nanofibres.

The surface morphology and size of the produced nanofibrous mats were characterized using a field emission scanning electron microscope (FESEM, AIS 2100, Korea). Besides, the size distributions of the nanofibres were analyzed using ImageJ software by randomly selecting 100 data points.

Differential scanning calorimetry (DSC) was performed using a Q600 (TA Instruments, USA) equipped with a N₂ atmosphere. The samples were heated from 25 °C to 600 °C at a heating rate of 10 °C/min. An

Table 1

Interpretation of diffusional release mechanisms of the PAO extract based on the value of “n”.

Release exponent (n)	Extract transport mechanism
n < 0.45	Pseudo-Fickian diffusion
0.45	Fickian diffusion
0.45 < n < 0.89	Non-Fickian (anomalous) transport
0.89	Case II (relaxational) transport
n > 0.89	Super case II transport

atomic force microscope (AFM, AFM/SPM Series 5500, Agilent) was used to analyze the surface roughness and topography of the electrospun membranes.

An ideal wound dressing should maintain an optimal humidity environment to increase the wound healing process. Wound dressings should have suitable water vapor permeability (WVP) and breathability to maintain vapor exchange at the wound. The WVP of nanofibres was determined according to a modified ASTM method [80,81]. Briefly, 7 mL of distilled water was placed in a penicillin bottle. Then, nanofibrous mats containing different concentrations of PAO extract were placed on the circular opening of the penicillin bottles to seal the opening. The bottles were placed in a desiccator at 25 °C with a relative humidity of 75% for 24 h. The WVP of the nanofibrous mats is defined as:

$$WVP (\%) = \frac{-\Delta\omega}{S} \times 100 \quad (1)$$

where $\Delta\omega$ is the change in water weight and S is the permeation area of the nanofibrous mats. Three samples were evaluated for each group.

2.4. Drug release mechanism

The in vitro release of the PAO extract from PVA/ALG nanofibres was monitored by UV spectrometry according to our previous work [79]. The cumulative amounts of released PAO from PVA/ALG nanofibres over time were evaluated, and then the kinetic model and mechanism of released PAO were investigated by fitting the obtained data to the proposed models [28,29]. The detail of the Korsmeyer-Peppas kinetic release equation model can be seen in the following:

$$\frac{M_t}{M_\infty} = kt^n \quad (2)$$

where M_t and M_∞ are the concentration of drug released at desired times, t, and the total concentration of PAO loaded in the nanofibres, respectively. Besides, k is the release rate constant related to the structural properties of the delivery system and the loaded substance. Moreover, n is the release exponent that illustrates the different release mechanisms of the PAO extract as described in Table 1.

The results of the in vitro release of the PAO extract from PVA/ALG nanofibres were plotted as $\log\left(\frac{M_t}{M_\infty}\right)$ versus the logarithm of time to investigate the release mechanism [79].

2.5. Cell culture and MTT assay

The biocompatibility of the electrospun nanofibres can be determined using in vitro bioassay methods such as the MTT assay. This test assesses how cytotoxic the wound dressing ingredients are against cultured cells. Therefore, the cytotoxicity of the PVA/ALG nanofibres with and without PAO loading was evaluated by the 3-(4,5-dimethyl-2-thiazolyl)-2,5-diphenyl-2H-tetrazolium bromide (MTT) assay. In brief, the L-929 cells (fibroblast cell line) were cultured in DMEM supplemented with 10% FBS, 1% penicillin/streptomycin, and 1% L-glutamine in an incubator set at 37 °C, 5% CO₂, and 95% relative humidity. Each nanofibrous mat sample (15 mm diameter) was exposed to UV irradiation for 30 min on each side. Then, they were soaked in DMEM and

seeded with cells in a 24-well plate at a density of 1×10^5 cells per well. Consequently, the cells were incubated at 37 °C for 1, 3, and 5 days. After that, the supernatant in each well was removed and replaced with 1 mL MTT solution. After 12 h of incubation at 37 ± 1 °C, 500 µL of DMSO was added to the wells to dissolve the formazan crystals. Finally, the absorbance of the dissolved solution was measured at 570 nm using a spectrophotometer.

In addition to the cytotoxicity assay mentioned above, cell attachment on the PVA/ALG nanofibres with and without PAO was graphically demonstrated. Briefly, cells were fixed to nanofibres using a 4% glutaraldehyde solution in PBS for 12 h. The morphology of the grown cells on nanofibrous mats was examined by SEM. Each group was evaluated three times.

2.6. Antibacterial studies

The agar diffusion technique is a relatively fast and easy-to-perform method to determine the antibacterial activity of diffusible antimicrobial components in the material. The antibacterial activity of the PVA/ALG nanofibres with and without PAO was investigated using the agar disk diffusion method against Gram-negative *Escherichia coli* (*E. coli*) and Gram-positive *Staphylococcus aureus* (*S. aureus*) bacteria. The Mueller-Hinton agar medium was obtained and sterilized at 121 °C for 17 min. Each agar plate was prepared by transferring 10 mL of agar medium into a Petri dish (100 mm × 15 mm) to be and solidified under sterile conditions in a laminar airflow hood for 1 h. Then, 1 mL of the bacteria suspension at a density of 10^6 CFU/mL was smeared over the agar surface, and the electrospun nanofibrous mats (2 cm diameter) were placed on the agar surface. Plates were then incubated at 37 ± 2 °C for 24 h. Finally, the antibacterial activity of the nanofibres was demonstrated by the diameter of the inhibition zone.

2.7. In vivo skin wound healing

2.7.1. Wound model and macroscopic measurements

Isogenic male albino mice weighing 25–30 g were used to evaluate skin wound healing of the produced PVA/ALG nanofibre scaffold under in vivo conditions. They were obtained from the Animal Centre of Bushehr University of Medical Sciences (BPUMS), Iran. All animals were treated according to generally accepted protocols for institutional animal care. Mice had free access to pellet food and water. For surgery, mice were anesthetized with an intraperitoneal injection of ketamine (60 mg/kg) and xylazine (10 mg/kg). Then they were shaved, and a wound of 1 cm in diameter was made on the back of each animal (in the interscapular region) with sharp surgical scissors under the conditions of surgical techniques.

A total of 18 animals were randomly divided into three experimental groups (n = 6 mice per group), which received topical treatment with normal saline (Group 1), the PVA/ALG nanofibrous mats loaded with PAO [PVA/ALG-PAO] (Group 2), and the PVA/ALG nanofibrous mats without PAO [PVA/ALG] (group 3). Immediately after making the skin excision, treatment with the indicated topical agent was started in each experimental group and repeated every 24 h. The wound beds were photographed after wound creation (day 0) and on days 3, 7, and 14 post-surgery to examine the wound area. The skin wound areas were determined by measuring the wound images using the ImageJ software. The following equation was used to calculate the wound area on the selected day concerning the initial size of that wound on day zero.

$$\text{Healing wound area (\%)} = \frac{A_u}{A_0} \times 100 \quad (3)$$

where A_u and A_0 are the wound areas on the understudy day and initial day (surgery day), respectively.

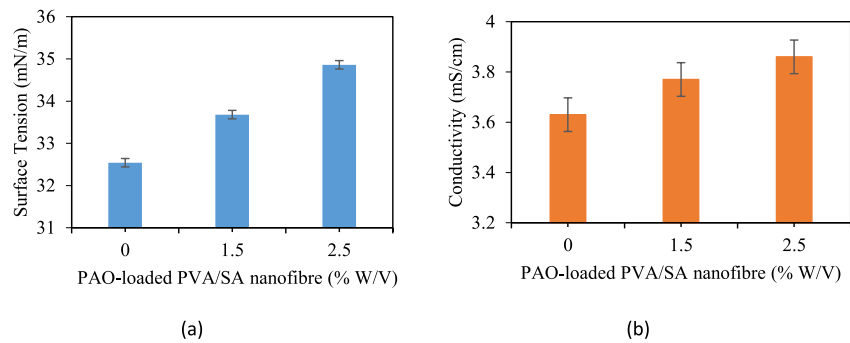


Fig. 1. (a) Surface tension and (b) conductivity of polymer solutions at the different concentrations of PAO.

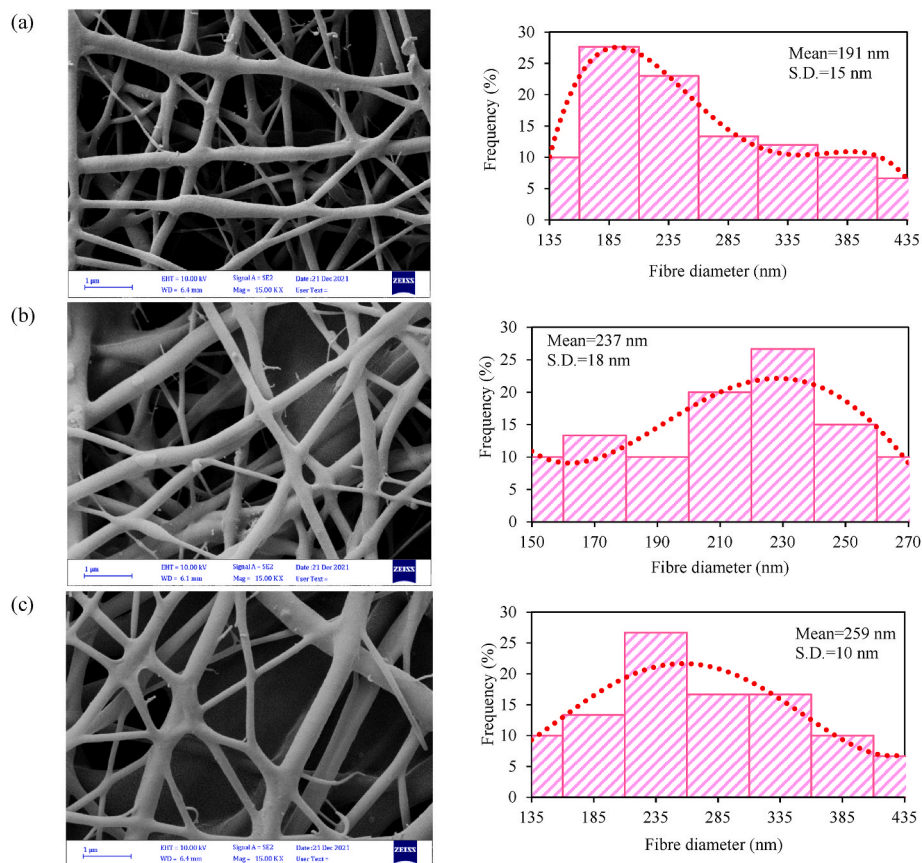


Fig. 2. SEM micrographs nanofibres morphology and corresponding histograms of their diameter distribution (a) PVA/ALG, (b) PVA/ALG-1.5% (w/v) PAO, and (c) PVA/ALG-2.5% (w/v) PAO.

2.7.2. Microscopic (histopathologic) evaluations

To assess histopathological variations, wound tissues were stained by the hematoxylin/eosin technique (H and E) on day 14 post-surgery. Briefly, cross sections of the wound tissue and surrounding intact skin were obtained. Thereafter, tissue samples were fixed by immersion in 3.7% formalin, embedded in paraffin, and 5 µm slices of them were stained according to the routine protocol of H and E. The two most important histopathological parameters assessed here were re-epithelization and gross inflammation in the wound tissues, which were qualitatively assessed by an expert pathologist and compared between groups.

2.7.3. Statistical analysis

Statistical analysis of in vivo skin wound healing was carried out by one-way analysis of variance (ANOVA) using post-hoc Tukey's method

test in SPSS® software version 21. Experimental wound area data are presented as mean wound area along with the calculated standard deviation. The statistical significance of the differences between the means was considered at $p < 0.05$.

3. Results and discussion

3.1. Physical properties of polymer solutions

The effect of the PAO concentration on the physical properties of the PVA/ALG/Triton solution, which is quite an important parameter in the electrospinning process, was investigated by varying it up to 2.5% (w/v) in the polymer solution. The surface tension and conductivity of polymer solutions, as a function of the amount of PAO, are shown in Fig. 1. The results showed that the surface tension and conductivity of blends were

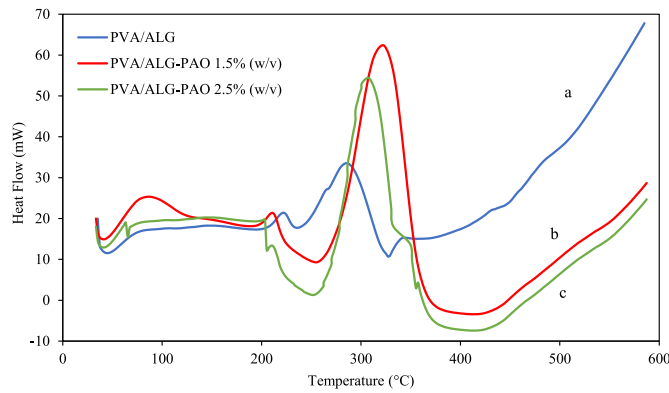


Fig. 3. Differential Scanning Calorimetry (DSC) thermograms of nanofibres: (a) PVA/ALG, (b) PVA/ALG-1.5% (w/v) PAO, and (c) PVA/ALG-2.5% (w/v) PAO.

Table 2

Thermal properties of PVA/ALG nanofibres by different concentrations of PAO extract.

Sample	T_m (°C)	ΔH_m (J/g)
PVA/ALG	286.3	42.878
PVA/ALG-1.5% (w/v) PAO	325.4	65.574
PVA/ALG-2.5% (w/v) PAO	310.6	57.250

increased after adding PAO extract. This could be due to the fact that the incorporation of a PAO into the polymer solution resulted in increased entanglement and/or chain-chain interaction between the PAO and the macromolecules of the polymer solution. The electrospinning process of polymer solutions is mainly influenced by surface tension and conductivity. Therefore, it can be concluded that increasing these factors could improve the electrospinnability of polymer solutions.

3.2. Physical properties of electrospun nanofibres

The performance of electrospun nanofibres in various fields has been affected by the morphological structure [82]. The engineering of high-performance nanofibres for proper drug release can be achieved by changing the structural composition of the surface and forming nanofibres without beads. Fig. 2 shows the morphology and size distribution of PVA/ALG nanofibres at different PAO concentrations. In all groups, the nanofibres were successfully fabricated and exhibited a uniform surface and were bead-free with no structural defects. Smooth PVA/ALG nanofibres; PAO-free nanofibres; with mean diameter sizes of $191 \text{ nm} \pm 15 \text{ nm}$ were successfully obtained. With the addition of PAO extract up to 1.5% (w/v) and 2.5% (w/v), the average diameter of the PVA/ALG nanofibres increased to $237 \text{ nm} \pm 18 \text{ nm}$ and $259 \text{ nm} \pm 10 \text{ nm}$, respectively. The PAO extract seems to be incorporated into PVA/ALG nanofibres as no PAO crystals were detected along the nanofibres. This finding clarified that the diameter of the nanofibres is directly related to

the polymer solution parameters such as surface tension and conductivity. An increase in the surface tension and conductivity of the feed-stock polymer solution, which was due to an increase in the PAO extract concentration, increased the mean diameter of the nanofibres.

To further investigate the thermal behavior of the produced nanofibres after adding different concentrations of PAO extract to the polymer solution, DSC thermograms were analyzed.

Fig. 3 shows the DSC curve of the electrospun PVA/ALG nanofibres containing different concentrations of the PAO extract. The samples were observed to have a peak at a temperature of 286.28°C , 325.4°C , and 310.6°C for PVA/ALG, PVA/ALG-1.5% (w/v) PAO, and PVA/ALG-2.5% (w/v) PAO relating to the melting points (T_m) of the samples, respectively.

Table 2 illustrates the thermal properties of the produced PVA/ALG nanofibres. The addition of PAO extract increased the thermal stability and enthalpy change of fusion (ΔH_m) of the PVA/ALG nanofibres; however, by increasing PAO extract concentration from 1.5% (w/v) to 2.5% (w/v) in the produced nanofibres, the T_m and ΔH_m values decreased from 325.4°C to 310.6°C and from 65.574 J/g to 57.25 J/g , respectively. All these results illustrated that PVA/ALG-1.5% (w/v) PAO nanofibres have higher thermal stability than PVA/ALG and PVA/ALG-2.5% (w/v) PAO nanofibres.

Table 3

Permeability properties of PVA/ALG nanofibres by different concentrations of PAO extract.

Sample	WVP ($\text{g}/\text{m}^2/\text{day}$)
PVA/ALG	2300
PVA/ALG-1.5% (w/v) PAO	2200
PVA/ALG-2.5% (w/v) PAO	2000

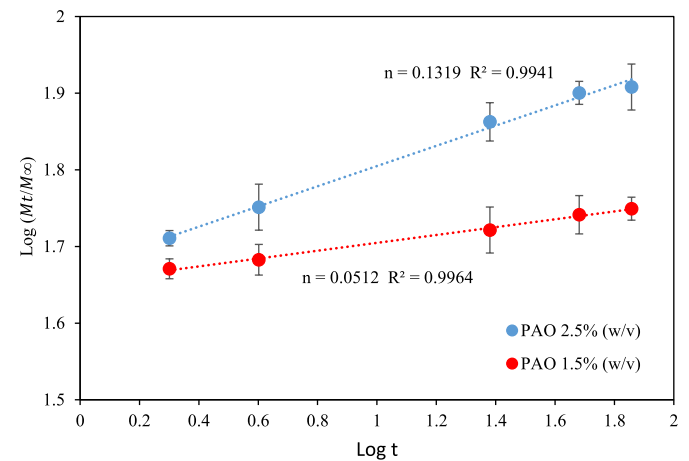


Fig. 5. Linear fits of the Korsmeyer-Peppas model for PAO release from nanofibres.

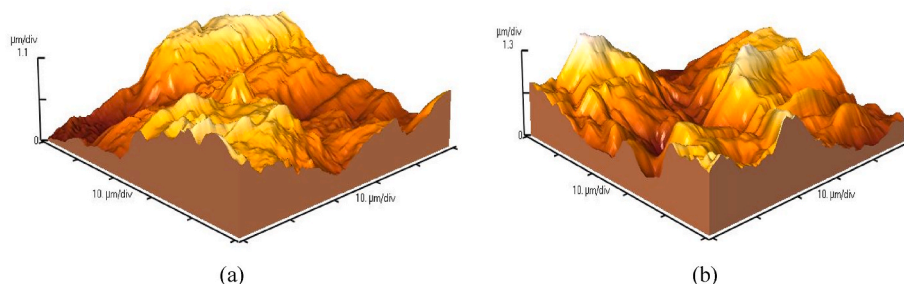


Fig. 4. Surface morphology and three-dimensional AFM images of PVA/ALG nanofibres (a) without and (b) with the PAO.

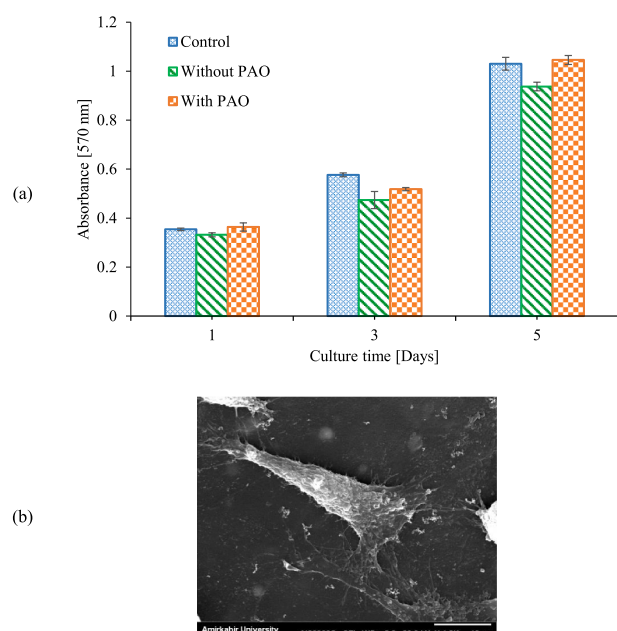


Fig. 6. (a) Evaluation of cytotoxic effects of PVA/ALG nanofibres with and without the PAO by MTT assay at 1, 3, and 5 days. Bars demonstrate mean value $\pm$ standard deviation ($n = 3$). (b) SEM micrographs show the morphology of fibroblast cells grown on the PVA/ALG-1.5% (w/v) PAO nanofibres after 72 h.

The atomic force microscope (AFM) can be used for 3D structure characterization and atomic manipulation on various surfaces. Therefore, the 3D AFM images and surface roughness of the electrospun PVA/

ALG nanofibres without and with PAO extract are shown in Fig. 4. This observation indicates that the addition of PAO extract to the nanofibres increases roughness compared to samples without PAO extract. The root mean square (RMS) roughness is obtained from the cross section of the mat surface. The RMS roughness of the PVA/ALG nanofibres without and with PAO extract was 48.72 nm and 41.2 nm, respectively, showing that the average roughness of the PVA/ALG nanofibres became lower and after adding the PAO extract decreased. This decrease in roughness can be attributed to the association of the PVA/ALG polymer solution and the PAO extract.

A proper wound dressing should have adequate WVP at an optimal rate to prevent excessive dehydration and buildup of exudates. The WVP rate for normal skin is around 204 g/m²/day. However, the water vapor loss of the injured skin should be in the range of 2000–2500 g/m²/day to avoid the risk of wound dehydration and to ensure adequate moisture levels [83]. Table 3 shows the corresponding WVP of PVA/ALG nanofibres at different concentrations of PAO extract. Compared to PVA/ALG nanofibres without PAO, all PAO-loaded nanofibres have a reducing effect on the WVP. Reducing the free volume of the nanofibre mat by PAO molecules can cause a reduction in WVP. The above results and the literature indicate that the WVP of all nanofibres is within the suitable range for an ideal wound dressing.

3.3. Drug release mechanism

To investigate the release kinetics, the Korsmeyer-Peppas model was used to develop the release profiles of the PAO extract from the nanofibres. Based on this model, the drug is released from a polymeric system by diffusion. The amounts of the Korsmeyer-Peppas slope or kinetic exponent (n) and the regression coefficient (R^2) are given in Fig. 5. The drug release profiles of PVA/ALG-1.5% (w/v) PAO and PVA/ALG-2.5% (w/v) PAO exhibited well agreement with the Korsmeyer-Peppas model

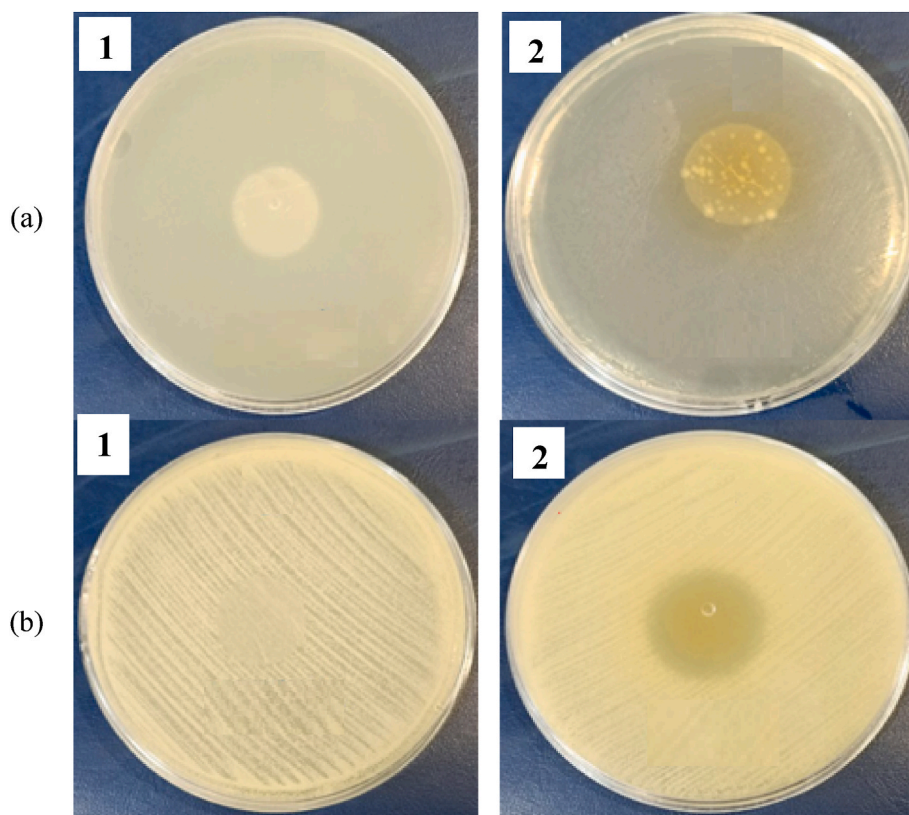


Fig. 7. Antibacterial activity results against (a) *E. coli* and (b) *S. aureus*. The numbers of the photos show (1) the PVA/ALG nanofibres without the PAO and (2) the PVA/ALG-1.5% (w/v) PAO nanofibres.

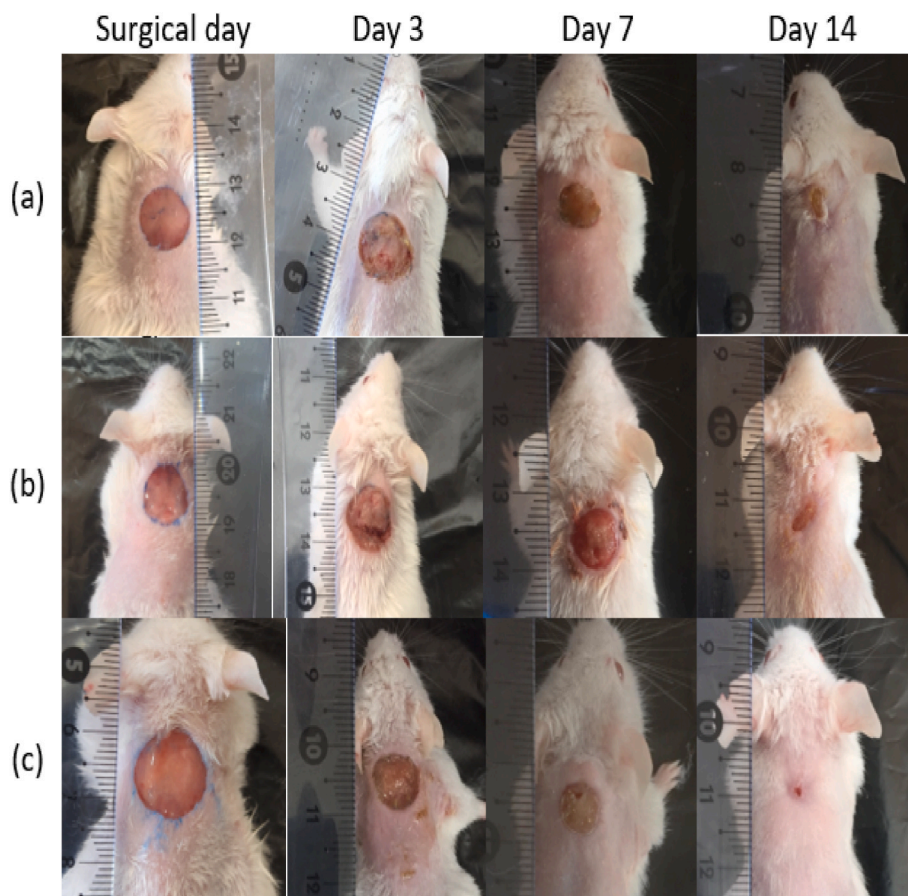


Fig. 8. The superficial wound healing process of full-thickness skin wounds treated with (a) normal saline, (b) PVA/ALG nanofibres, and (c) PVA/ALG-1.5% (w/v) PAO nanofibres on days 0, 3, 7 and 14 post-surgery ($n = 4-6$ mice per group).

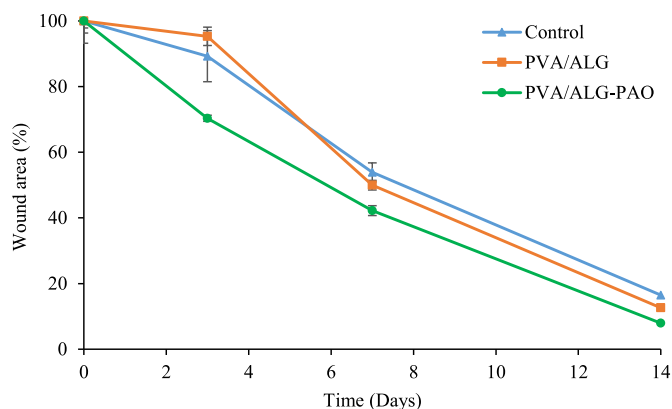


Fig. 9. Variation in wound area (mean $\pm$ standard deviation) with time within the various treatment groups (a) normal saline (control), (b) PVA/ALG nanofibres, and (c) PVA/ALG-1.5% (w/v) PAO nanofibres ($n = 4-6$ mice per group).

($R^2 = 0.99$). Based on the results, the adjusted values of n for both fabricated nanofibres containing PAO extract are less than 0.45. Thus, regarding the data in Table 1, the release mechanism of the PAO extract from the electrospun nanofibres followed the pseudo-Fickian diffusion. It can be asserted that PAO release from PVA/ALG nanofibres in all concentrations can be analyzed and controlled by the Korsmeyer-Peppas model and the pseudo-Fickian mechanism.

3.4. Cell culture and MTT assay

The level of cytotoxicity of the electrospun PVA/ALG nanofibres without and with PAO extract was evaluated for different periods of time in terms of fibroblasts cell viability using the MTT cytotoxicity assay after incubation of the nanofibres with the fibroblast cells, and the obtained results are shown in Fig. 6-a. These results show that OD or viability of fibroblast cells are not significantly different and are similar for all groups after one day of cell seeding. Besides, the viability of the cultured cells enhanced over time for all groups. After 3 and 5 days, the OD values in the PVA/ALG-PAO nanofibres and the control groups do not illustrate statistical significance, which admitted the lack of any cytotoxic effects of the electrospun nanofibres. However, the OD value of PVA/ALG nanofibres containing PAO extract is much higher than that of the pure PVA/ALG nanofibres five days after incubation.

The morphology of fibroblast cells growing on the surface of the PVA/ALG-1.5% (w/v) PAO nanofibres is shown by SEM images in Fig. 6-b. After 72 h of culture, fibroblast cells can be seen attached to the PVA/ALG-1.5% (w/v) PAO nanofibres. They were fully extended and altered their round shape to form pseudopods, suggesting that the fibroblast cells can penetrate within the electrospun nanofibre mats. This observation of cellular morphology was consistent with the results of the MTT assay.

The findings concluded that the fabricated PVA/ALG-PAO nanofibres are not only non-toxic to fibroblast cells, but also provide better environments to promote cell survival and attachment.

3.5. Antibacterial studies

The antibacterial activities of PVA/ALG nanofibres with and without

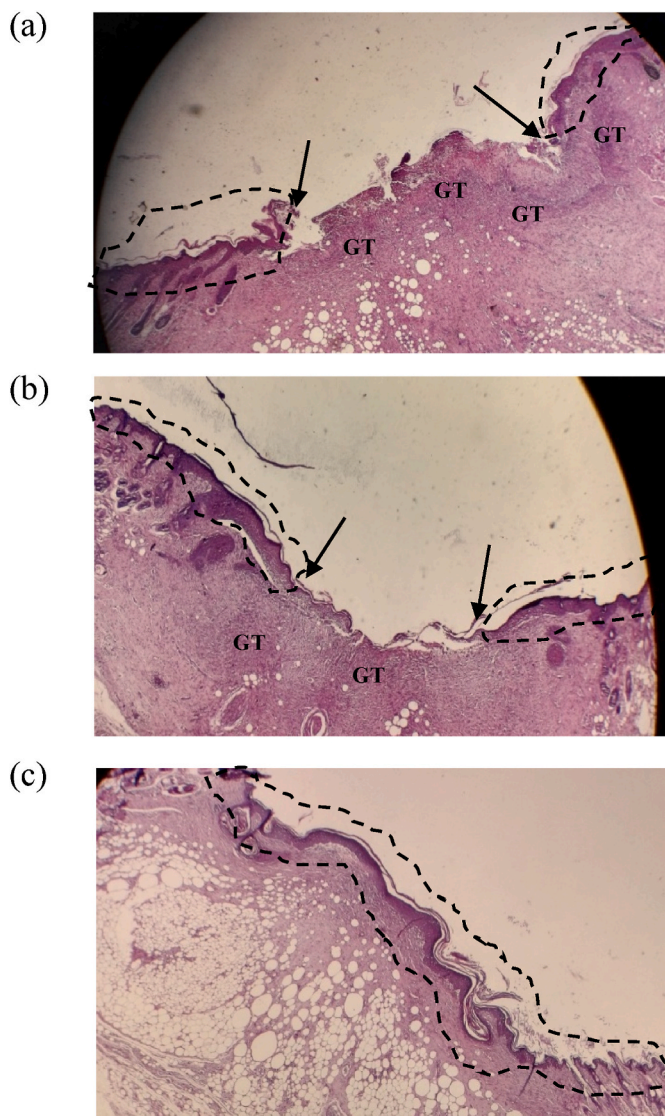


Fig. 10. Histological findings of the wound tissue on day 14 post-surgery treated with (a) normal saline, (b) PVA/ALG nanofibres, and (c) PVA/ALG-1.5% (w/v) PAO nanofibres. Arrows (→) demonstrate wound re-epithelialization edges; GT stands for granulation tissue.

PAO against *S. aureus* (Gram-positive) and *E. coli* (Gram-negative) were evaluated using the agar disk diffusion method (Fig. 7). PVA/ALG nanofibres without PAO served as a control and no zone of inhibition was distinguished around them against both strains as marked by region number 1 in Fig. 7-a and 7-b. PVA/ALG-1.5% (w/v) PAO nanofibres illustrated different antibacterial activities against the tested Gram-negative and Gram-positive bacteria as shown in region number 2 of Fig. 7-a and 7-b. The PVA/ALG-1.5% (w/v) PAO nanofibres showed a zone of inhibition of $5 \text{ mm} \pm 0.5 \text{ mm}$ against *S. aureus* compared with no zone of inhibition against *E. coli*. Therefore, PVA/ALG-1.5% (w/v) PAO nanofibres proved to be more effective on *S. aureus* as a Gram-positive bacterium which is involved in many infected wounds [84].

3.6. In vivo results

3.6.1. Macroscopic analyses

Fig. 8 shows the superficial wound healing process of full-thickness wounds treated with normal saline, PVA/ALG nanofibres, and PVA/ALG-1.5% (w/v) PAO nanofibres at different time points after surgery. Besides, changes in the wound areas over time are shown in Fig. 9. The

wound areas of all groups narrow from day 0 to day 14. However, the rate of wound healing differs between groups. For example, the fastest wound closure was observed in the wounds treated with PVA/ALG-1.5% (w/v) PAO nanofibres with an average wound area of $7.9\% \pm 1.5\%$ on day 14 after surgery. The slowest rate was in the normal saline treated wounds, which had an average wound area of $16.48\% \pm 2.89\%$ after 14 days of treatment.

One-way ANOVA test illustrated a significant difference between the mean areas of the groups at day 3 ($F = 31.763$, $df = 2$, $p < 0.001$), at day 7 ($F = 6.093$, $df = 2$, $p = 0.021$), and at day 14 after surgery ($F = 17.219$, $df = 2$, $p = 0.001$). On day 3 post-surgery, the post-hoc test shows that the PVA/ALG-1.5% (w/v) PAO nanofibres resulted in smaller wound areas ($70.35\% \pm 3.68\%$) than the PVA/ALG nanofibres ($95.32\% \pm 2.13\%$) ($p < 0.001$) and normal saline solution ($89.28\% \pm 6.8\%$) ($p = 0.001$). At day 7 post-surgery, the PVA/ALG-1.5% (w/v) PAO nanofibres generate more contracted wounds ($42.22\% \pm 0.93\%$), compared to normal saline ($49.97\% \pm 2.80\%$) ($p = 0.019$). After 14 days of surgery, the PVA/ALG-1.5% (w/v) PAO nanofibres are again associated with significantly smaller wounds ($7.93\% \pm 1.50\%$) than the PVA/ALG nanofibres ($12.61\% \pm 1.46\%$) ($p = 0.026$) and normal saline ($16.48\% \pm 2.89\%$) ($p = 0.001$). In summary, the results illustrate that loading PAO extract into PVA/ALG nanofibres can accelerate skin repair and healing processes.

3.6.2. Microscopic analyses

For a deeper description of the wound healing process, photomicrographs of H and E-stained wound tissue after 14 days of treatment are shown in Fig. 10. The normal saline group showed very slow re-epithelialization of the wound even after 14 days. The wound stroma in this group was highly edematous with vivid granulation tissue that had not resolved even after 14 days (Fig. 10-a). Granulation tissue consists of inflammatory cells, activated fibroblasts, neovascularity, etc., which is essential at the beginning of the wound healing process. However, the presence of granulation tissue at later stages (i.e., day 14 post-surgery) signifies a delay or defect in the wound healing process. Fig. 10-b shows the degree of wound healing after 14 days of treatment with PVA/ALG nanofibres. The wound treated with these nanofibres exhibited less inflammatory infiltration and greater re-epithelialization compared to the normal saline group. Wounds treated with PVA/ALG-1.5% (w/v) PAO nanofibres presented a faster healing rate and better quality in terms of more complete re-epithelialization and almost no inflammation (granulation tissue) than the other groups (Fig. 10-c). In other words, the new epidermal layer was intact and covered with keratinization, and the wound stroma was filled with a fibrous context rather than granulation tissue. These results confirmed the macroscopic findings of wound healing and illustrated that PVA/ALG nanofibres loaded with PAO extract provided a superior physiological environment promote wound healing compared to other treatments.

4. Conclusions

Wound healing is a complex and dynamic process that involves replacing damaged tissue with newly produced tissue layers. Wound dressings containing natural biomaterials have attracted more attention due to their biodegradability, biocompatibility, effectiveness in wound healing, appropriate antibacterial properties, and renewable properties. Therefore, this study evaluated the effects of adding herbal medicinal product from PAO extract to PVA/ALG polymer solution, including surface tension and conductivity of polymer solutions, morphology, surface roughness, thermal properties, water vapor permeability (WVP), antibacterial activity, and in vitro/in vivo efficacy of the fabricated nanofibres for wound healing. The results showed that increasing the PAO extract concentration to 1.5% (w/v) as the optimal composition increased the surface tension and conductivity of polymer solutions by 3.5% and 3.86%, respectively, and increased the diameter of nanofibres by $191 \text{ nm} \pm 15 \text{ nm}$ – $259 \text{ nm} \pm 10 \text{ nm}$. The chain-chain interaction

could be the reason for this increase in surface tension and conductivity of the polymer solutions and the diameter of the nanofibres. The AFM results also illustrated that the surfaces of the PVA/ALG nanofibres without PAO were 18.25% rougher than those of the PVA/ALG nanofibres with PAO. This result demonstrates that the PAO formed internal and uniform bonds with the PVA/ALG solution, creating a more homogeneous polymer solution and reducing the roughness of the mat surface. On the other hand, the DSC analyses indicated that adding the PAO to the PVA/ALG polymer solution by up to 1.5%, increases the thermal resistance and improves the thermal behavior and heat stability of the produced nanofibres due to a strong interface interaction between PAO and PVA/ALG polymer solution. Additionally, the fabricated nanofibres provide an ideal wound healing system with great properties such as creating an antibacterial and moist environment while allowing oxygen exchange. The in vitro release profiles of the PAO showed that the drug release from the PVA/ALG nanofibres follows the Korsmeyer-Peppas model and the pseudo-Fickian diffusion mechanism. Furthermore, the in vitro MTT cell viability assay and cell culture results demonstrated that the PVA/ALG-1.5% (w/v) PAO nanofibres were capable of promoting cell adhesion, fibroblast cells attachment, and cell proliferation without any cytotoxicity. Finally, in vivo experiments showed that the PVA/ALG-1.5% (w/v) PAO nanofibres had a higher wound healing rate (92.07%) and a greater reduction in wound inflammation compared to the PVA/ALG nanofibres delivered. In conclusion, the electrospun PVA/ALG-1.5% (w/v) PAO nanofibres could be a promising candidate for wound healing and tissue engineering

Nomenclature

A_0	Wound areas on the initial day (surgery day) [mm ²]
A_u	Wound areas on the understudy day [mm ²]
ALG	Sodium alginate
ΔH_m	Enthalpy change of fusion [J/g]
k	Release rate constant
M_t	The concentration of drug released at desired times, t
M_∞	The total concentration of the PAO-loaded in the nanofibres
N	Release exponent
PAO	<i>Pistacia atlantica</i> oil
PVA	Polyvinyl alcohol
T_m	The temperature of fusion [°C]

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applications.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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